

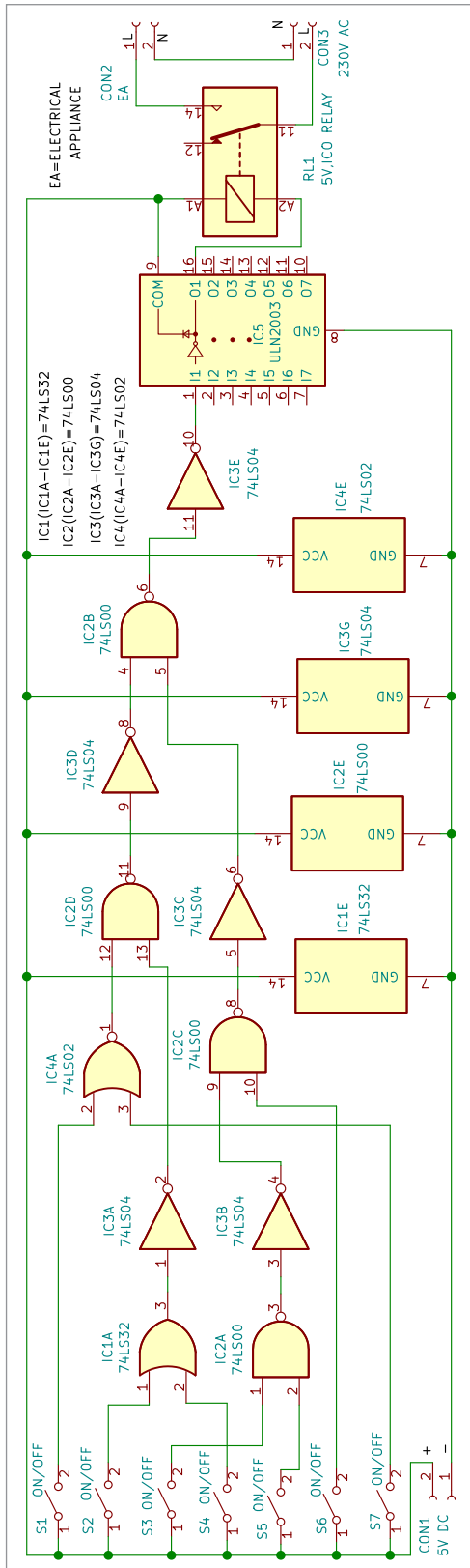


Fig. 2: Circuit design

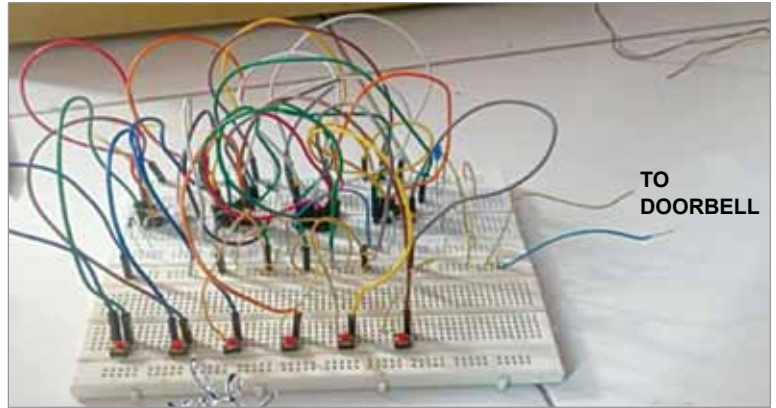


Fig. 3: Author's prototype

significantly reduces the occurrence of bothersome and unexpected doorbell disturbances.

Setting the password is a straightforward process. The password is configured using switches S1 through S7. When a switch is open, it signifies a code of 0; when closed, it represents a code of 1.

In this instance, our password is 0010110, which corresponds to the configuration of switches S1 through S7.

Based on this code the final output of logic gates using IC1 through IC4 will be 5V high (logic 1). This high output is further given to the input pin1 of relay driver IC ULN2003 and drives the relay. So the connected load will come in contact with power supply and switch on.

Construction and testing

While the circuit is based on logic gates and lacks the capacity for password modification, a PCB design is not necessary. As demonstrated by the author's prototype in Fig. 3, the project can be assembled on a breadboard or

a general-purpose PCB. To activate the connected electrical appliance (here a 60W bulb), input the correct password sequence (0010110) using switches S1 through S7. If the code is incorrect, the load will remain inactive.

We have used the ULN2003 IC and an electric bulb as a load. For alternative applications, the ULN2003 IC and an electric bulb can be replaced with a doorbell mechanism to create a password-based doorbell.

Password-based electrical appliance control offers a refreshing perspective on household convenience and security. With a straightforward circuit design and a robust working principle, this project showcases the amalgamation of innovation and practicality. Whether for appliance control or doorbell security, this system offers a cost-effective and elegant solution.

EFY note. The password cannot be changed as it is logic gate based project. **EFY**

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